Yannik Schuler

Algebraic Geometer & Mathematical Physicist

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Employment

Sep 2024 – Dec 2027 **Postdoc**
ETH Zürich, Department of Mathematics
Funded partly by a Walter Benjamin Fellowship of the DFG (Jan 2026–Dec 2027)
In Rahul Pandharipande's research group

Mar 2024 – Aug 2024 **Research Associate**
University of Cambridge, DPMMS
In Dhruv Ranganathan's research group

Education

Nov 2020 – Jun 2024 **PhD Mathematics**
University of Sheffield
Supervisor: Andrea Brini
Thesis: "Topics in Gromov–Witten theory"

Apr 2017 – Jun 2020 **MSc Physics**
University of Bonn
Thesis: "Higher quantum Airy structures in two-dimensional topological gravity"
under supervision of Albrecht Klemm and Gaëtan Borot

Apr 2017 – Mar 2019 **BSc Mathematics**
University of Bonn
Thesis: "Representations of modulated quivers"
under supervision of Jan Schröer

Oct 2013 – Sep 2016 **BSc in Physics**
University of Bonn
Thesis: "Gravitational waves in view of the discovery at LIGO"
under supervision of Albrecht Klemm

Publications

Papers

- Michel van Garrel, Navid Nabijou, and Yannik Schuler. "Gromov–Witten theory of bicyclic pairs". *Trans. Am. Math. Soc.* 379.5 (2026), pp. 3353–3402. DOI: 10.1090/tran/9581.
- Yannik Schuler. "The log-open correspondence for two-component Looijenga pairs". *Forum Math. Sigma* 13 (2025). Id./No. e102, p. 30. DOI: 10.1017/fms.2025.10042.
- Andrea Brini and Yannik Schuler. "On quasi-tame Looijenga pairs". *Commun. Num. Theor. Phys.* 17.2 (2023), pp. 313–341. DOI: 10.4310/CNTP.2023.v17.n2.a3.
- Gaëtan Borot, Reinier Kramer, and Yannik Schüler. "Higher Airy structures and topological recursion for singular spectral curves". *Ann. Inst. Henri Poincaré Comb. Phys. Interact.* (2023). DOI: 10.4171/AIHPD/168.

Pre-Prints

- A. Dänzer and Y. Schuler. "Gromov-Witten theory of the local model of the standard flop". 2026. arXiv: 2611.01260 [math.AG].
- Daniel Holmes and Yannik Schuler. "Experiments with membranes, maps and sheaves". 2026. arXiv: 2609.03152 [math.AG].
- Alessandro Giacchetto, Rahul Pandharipande, and Yannik Schuler. "Membranes and Maps". 2026. arXiv: 2609.02409 [math.AG].
- Yannik Schuler. "Gromov–Witten invariants and membrane indices of fivefolds via the topological vertex". 2026. arXiv: 2603.24256 [math.AG].
- Andrea Brini and Yannik Schuler. "Refined Gromov–Witten invariants". 2024. arXiv: 2410.00118 [math.AG].

Talks & Presentations

Conferences & workshops

Jul 2026 *Membranes and Maps*, Conference "Refinements in Enumerative Geometry and Physics" (ICTS Bengaluru, IN)

May 2026 *Fivefold perspective on the GW/DT correspondence*, Conference "Hilbert in Mirror" (SIMIS Shanghai, CN)

Jun 2025 *Curves in 5-folds and conjectures on Hodge integrals*, Conference "Helvetic Algebraic Geometry Seminar" (Les Diablerets, CH)

Jan 2025 *Membranes and Maps*, Conference "Algebra and Quantum Geometry of BPS quivers" (Les Diablerets, CH)

Sep 2024 *Maps, Sheaves and Membranes*, Conference "Duality, Geometry and Physics" (Bernoulli center, Lausanne, CH)

Sep 2024 *A worldsheet definition of the refined topological string*, Conference "Geometry and Physics of Mirror Symmetry" (University of Sheffield, UK)

May 2024 *Gromov–Witten theory and the refined topological string*, Conference "Real and Logarithmic Enumerative Geometry" (Oberwolfach, DE)

Jul 2022 *The Log-Local-Open Correspondence*, Program "K-theory, algebraic cycles and motivic homotopy theory" (INI Cambridge, UK)

Apr 2022 *Correspondences in enumerative geometry*, Student talk at "LMS Regional Meeting" (University of Birmingham, UK)

Seminars

May 2026 *Membranes and Maps*, Heidelberg Algebraic Geometry Seminar (University of Heidelberg, DE)

Nov 2025 *Equivariant topological strings on fivefolds*, Physical Mathematics Seminars (University of Geneva, CH)

Oct 2025 *Stable maps to fivefolds*, Harvard–MIT AG Seminar (MIT, US)

Oct 2025 *Refined Gromov–Witten invariants*, Geometry, Symmetry, and Physics Seminar (Yale University, US)

Oct 2025 *Membranes and Maps*, Informal Mathematical Physics Seminar (Columbia University, US)

Oct 2025 *Quintuple Hodge Integrals*, Algebraic Geometry and Moduli Seminar (ETH Zürich, CH)

Aug 2025 *Refined Gromov–Witten invariants*, Moscow-Beijing topology seminar (online)

Oct 2024 *Refined mirror symmetry of local* $\mathbb{P}^2$, Algebraic Geometry and Moduli Seminar (ETH Zürich, CH)

May 2024 *Log vs Open Gromov–Witten theory*, Fanosearch Seminar (Imperial College, UK)

May 2024 *Tangents to the nodal cubic*, Calf meeting (University of Bath, UK)

Apr 2024 *Tangents to the nodal cubic*, Junior Geometry Seminar (University of Cambridge, UK)

Apr 2024 *Gromov–Witten theory and the refined topological string*, Algebra Seminar (University of Edinburgh, UK)

Dec 2023 *Gromov–Witten theory and the refined topological string*, Algebraic Geometry and Moduli Seminar (ETH Zürich, CH)

Nov 2023 *Gromov–Witten theory from the fivefold perspective*, Mathematical Physics Seminar (HU Berlin, DE)

Nov 2023 *Equivariant Strings and Gromov–Witten theory*, Algebraic Geometry Seminar (Utrecht University, NL)

Nov 2023 *Equivariant Curves and Strings*, MAGIC seminar (Imperial College, UK)

Oct 2023 *Tangents to the Nodal Cubic*, Geometry and Physics Seminar (University of Sheffield, UK)

May 2023 *Gromov–Witten theory from the fivefold perspective*, Geometry and Mathematical Physics Seminar (University of Birmingham, UK)

Feb 2023 *The Log-Local-Open Correspondence*, Algebraic Geometry Seminar (Cambridge University, UK)

Teaching Experience

Sep 2025 – **Lecturer**
ETH Zürich
- Bachelor Seminar "Quiver Representations" (Fall 2025)
- Master Seminar "Gromov–Witten theory" (Fall 2026)

Feb 2022 – Jun 2023 **Postgraduate Teaching Assistant**
University of Sheffield
- Demonstrator & Marker for "Core Foundation Mathematics" (Prep Course)
- Demonstrator & Marker for "Mathematical modelling" (BSc)
- Demonstrator & Marker for "Foundations of Pure Mathematics" (BSc)
- Marker for "Advanced Calculus and Linear Algebra" (BSc)

Apr 2015 – Sep 2020 **Teaching Assistant**
University of Bonn
- Demonstrator & Marker for "Mathematics for physicists" (BSc)
- Demonstrator & Marker for "Mathematics for computer scientists" (BSc)
- Demonstrator & Marker for "Theoretical classical mechanics" (BSc)
- Demonstrator & Marker for "Theoretical electrodynamics" (BSc)
- Demonstrator & Marker for "Theoretical statistical physics" (BSc)
- Head Tutor for "General Relativity" (MSc)

Student supervision

Master theses

- A. Dänzer, Fall 2025, *GW theory of a higher-dimensional generalization of the resolved conifold*, arXiv:2611.01260

Bachelor theses

- O. Galy, Fall 2026 (ongoing), *Gabriel's theorem via Coxeter functors*
- J. Corti, Spring 2026, *The McKay Correspondence*
- A. Kalman, Fall 2025, *Burnside's Character Formula for Hurwitz Numbers*

Semester papers

- A. Kalman, Fall 2026 (ongoing), *$SU(2)$ gauge theory on $\mathbb{C}^2$ vs. GW theory on local $\mathbb{P}^1 \times \mathbb{P}^1$*
- E. Dikhoff, Fall 2026 (ongoing), *Hurwitz Numbers*

Academic service

Referee Full reports for *Annales Henri Poincaré*, *Communications in Mathematical Physics*, *International Mathematics Research Notices*, *Journal of Algebraic Geometry*, *Transactions of the American Mathematical Society*
Quick opinions for *Advances in Mathematics*, *Journal de l'École polytechnique Mathématiques*

Reviewer for *zbMATH Open* and the AI mathematics benchmark *IMProofBench*

Scholarships

- Walter Benjamin Fellowship of the Deutsche Forschungsgemeinschaft (Jan 2026 – Dec 2027)
- Scholarship of the Bonn-Cologne Grad. School of Physics and Astronomy (Oct 2017 – Sep 2019)
- University of Sheffield Faculty of Science Scholarship (Nov 2020 – May 2024)